

TotalWaitingTime

There are N clients who have ordered N handmade items. The K -th client ordered exactly one item that takes $T[K]$ hours to make. There is only one employee who makes items for clients, and they work in the following manner:

- spend one hour making the first item;
- if the item is finished, the employee delivers it to the client immediately;
- if the item is not finished, they put it after the N -th item for further work;
- the employee starts making the next item.

For example for $T = [3, 1, 2]$, the employee spends 6 hours making items in the following order: [1, 2, 3, 1, 3, 1]. The first client waited 6 hours for their item, the second client received their item after 2 hours and the third client after 5 hours. What is the total time that clients need to wait for all ordered items? For the above example, the answer is $6 + 2 + 5 = 13$.

As the result may be large, return its last nine digits without leading zeros (in other words, return the result modulo 10^9).

Write a function:

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def solution(T)
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that, given an array of integers T of length N , returns the total time that the clients need to wait (modulo 10^9).

Examples:

1. Given $T = [3, 1, 2]$, the function should return 13 as explained above.
2. Given $T = [1, 2, 3, 4]$, the function should return 24. The employee prepares the items in the following order: 1, 2, 3, 4, 2, 3, 4, 3, 4, 4. The first client waited for 1 hour, the second client for 5 hours, the third client for 8 hours, and the fourth client for 10 hours. The total waiting time of all clients is $1 + 5 + 8 + 10 = 24$ hours.
3. Given $T = [7, 7, 7]$, the function should return 60.
4. Given $T = [10000]$, the function should return 10000.

Write an efficient algorithm for the following assumptions:

- N is an integer within the range $[1..100,000]$;
- each element of array T is an integer within the range $[1..10,000]$.